THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 31	Add the two distances from the centre.
2	**(d)** 17	Subtract the two distances from the centre.
3	**(b)** 14	Add the distances of the centres from the common chord.
4	**(c)** 60°	Add the two parts of the angle.
5	**(d)** 108°	Linear pair at D.
6	**(a)** 132°	Opposite angles of a cyclic quadrilateral.
7	**(d)** 168	Area $= \frac{1}{2} \times$ chord $\times$ distance.
8	**(a)** both diameters	Reason with circle theorems.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Half-chord $= \sqrt{5^2 - 3^2} = 4$ cm, using the perpendicular bisection in R; chord $= 8$ cm.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true theorems, but R does not by itself explain why equal chords are equidistant.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distances: $\sqrt{25^2 - 7^2} = 24$ cm and $\sqrt{25^2 - 24^2} = 7$ cm (1); Distance $= 24 + 7 = 31$ cm (1).	1+1
12	Distances of centres from the chord: 6 cm and 15 cm (1); Distance $= 21$ cm (1).	1+1
13	$\angle ABC = 32^\circ + 41^\circ = 73^\circ$ (1); $\angle ADC = 180^\circ - 73^\circ = 107^\circ$ (1).	1+1
14	Distances: $\sqrt{25^2 - 15^2} = 20$ cm and $\sqrt{25^2 - 20^2} = 15$ cm (1); Distance $= 20 + 15 = 35$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$d_1 = \sqrt{13^2 - 12^2} = 5$ cm, $d_2 = \sqrt{15^2 - 12^2} = 9$ cm (1); Opposite sides: $5 + 9 = 14$ cm (1); Same side: $\lvert 5 - 9 \rvert = 4$ cm (1).	1+1+1
16	Distances from the centre 12 cm and 5 cm (1); Same side: 7 cm (1); Opposite sides: 17 cm (1).	1+1+1
17	$\angle BCD = 180^\circ - 34^\circ = 146^\circ$ (1); $\angle ADB = 90^\circ$ (angle in a semicircle) (1); $\angle ABD = 90^\circ - 34^\circ = 56^\circ$ (1).	1+1+1
	OR: $d_1 = \sqrt{13^2 - 12^2} = 5$ cm, $d_2 = \sqrt{20^2 - 12^2} = 16$ cm (1); Opposite sides: $5 + 16 = 21$ cm (1); Same side: $\lvert 5 - 16 \rvert = 11$ cm (1).	1+1+1
18	Distances from the centre 15 cm and 8 cm (1); Same side: 7 cm (1); Opposite sides: 23 cm (1).	1+1+1
	OR: $\angle BCD = 180^\circ - 41^\circ = 139^\circ$ (1); $\angle ADB = 90^\circ$ (angle in a semicircle) (1); $\angle ABD = 90^\circ - 41^\circ = 49^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The perpendicular from the centre bisects the chord, so the half-length is 12 m (1). (ii) $\sqrt{13^2 - 12^2} = 5$ m (1). (iii) Half-chord $= \sqrt{13^2 - 12^2} = 5$, so the chord is 10 m (1). (iv) Distances from the centre are 5 m and 12 m on opposite sides, so the distance between them is $5 + 12 = 17$ m (2).	1+1+1+2
20	$\angle A + \angle C = 180^\circ$ (1); $5x = 160$, so $x = 32$ (1); $\angle B + \angle D = 180^\circ$ (1); $6y = 162$, so $y = 27$ (1); $\angle A = 101^\circ$, $\angle C = 79^\circ$, $\angle B = 87^\circ$, $\angle D = 93^\circ$ (1).	1+1+1+1+1
	OR: Half-chord $= 9$ cm, $d = \sqrt{15^2 - 9^2} = 12$ cm (1); Area $= \frac{1}{2} \times 18 \times 12 = 108$ cm^2 (1); $OA = OB = 15$ cm (1); Perimeter $= 15 + 15 + 18 = 48$ cm (1); Distance, area and perimeter obtained (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).

- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).
- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 21	Add the distances of the centres from the common chord.
2	**(b)** 80°	Add the two parts of the angle.
3	**(a)** 95°	Linear pair at D.
4	**(a)** 123°	Opposite angles of a cyclic quadrilateral.
5	**(c)** 300	Area $= \frac{1}{2} \times$ chord $\times$ distance.
6	**(d)** rectangle	Reason with circle theorems.
7	**(b)** 14	Add the two distances from the centre.
8	**(b)** 2	Subtract the two distances from the centre.
9	**(c)** A is true but R is false.	A is the cyclic-quadrilateral theorem; R is false in general (e.g. angles 80° and 100°).
10	**(d)** A is false but R is true.	A is false because every chord is at most as long as the diameter; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distances of centres from the chord: 8 cm and 20 cm (1); Distance $= 28$ cm (1).	1+1
12	$\angle ABC = 27^\circ + 49^\circ = 76^\circ$ (1); $\angle ADC = 180^\circ - 76^\circ = 104^\circ$ (1).	1+1
13	Distances: $\sqrt{10^2 - 6^2} = 8$ cm and $\sqrt{10^2 - 8^2} = 6$ cm (1); Distance $= 8 + 6 = 14$ cm (1).	1+1
14	Distances of centres from the chord: 15 cm and 21 cm (1); Distance $= 36$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$d_1 = \sqrt{25^2 - 24^2} = 7$ cm, $d_2 = \sqrt{26^2 - 24^2} = 10$ cm (1); Opposite sides: $7 + 10 = 17$ cm (1); Same side: $\lvert 7 - 10 \rvert = 3$ cm (1).	1+1+1
16	Distances from the centre 24 cm and 7 cm (1); Same side: 17 cm (1); Opposite sides: 31 cm (1).	1+1+1
17	$\angle BCD = 180^\circ - 52^\circ = 128^\circ$ (1); $\angle ADB = 90^\circ$ (angle in a semicircle) (1); $\angle ABD = 90^\circ - 52^\circ = 38^\circ$ (1).	1+1+1
	OR: $d_1 = \sqrt{25^2 - 24^2} = 7$ cm, $d_2 = \sqrt{30^2 - 24^2} = 18$ cm (1); Opposite sides: $7 + 18 = 25$ cm (1); Same side: $\lvert 7 - 18 \rvert = 11$ cm (1).	1+1+1
18	Distances from the centre 20 cm and 15 cm (1); Same side: 5 cm (1); Opposite sides: 35 cm (1).	1+1+1
	OR: $\angle BCD = 180^\circ - 63^\circ = 117^\circ$ (1); $\angle ADB = 90^\circ$ (angle in a semicircle) (1); $\angle ABD = 90^\circ - 63^\circ = 27^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\angle ACB = \frac{1}{2} \times 110^\circ = 55^\circ$ (1). (ii) 90° (angle in a semicircle) (1). (iii) Angles in the same segment are equal, so $\angle ADB = 55^\circ$ (1). (iv) ACBE is a cyclic quadrilateral, so $\angle AEB = 180^\circ - 55^\circ = 125^\circ$ (2).	1+1+1+2
20	$\angle A + \angle C = 180^\circ$ (1); $6x = 150$, so $x = 25$ (1); $\angle B + \angle D = 180^\circ$ (1); $6y = 162$, so $y = 27$ (1); $\angle A = 70^\circ$, $\angle C = 110^\circ$, $\angle B = 66^\circ$, $\angle D = 114^\circ$ (1).	1+1+1+1+1
	OR: Half-chord $= 7$ cm, $d = \sqrt{25^2 - 7^2} = 24$ cm (1); Area $= \frac{1}{2} \times 14 \times 24 = 168$ cm^2 (1); $OA = OB = 25$ cm (1); Perimeter $= 25 + 25 + 14 = 64$ cm (1); Distance, area and perimeter obtained (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).
- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 82°	Linear pair at D.
2	**(a)** 114°	Opposite angles of a cyclic quadrilateral.
3	**(c)** 240	Area $= \frac{1}{2} \times$ chord $\times$ distance.
4	**(b)** square	Reason with circle theorems.
5	**(c)** 7	Add the two distances from the centre.
6	**(c)** 1	Subtract the two distances from the centre.
7	**(c)** 25	Add the distances of the centres from the common chord.
8	**(a)** 80°	Add the two parts of the angle.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{1}{2} \times 100^\circ = 50^\circ$, directly from R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true theorems, but R is a special case about semicircles and does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle ABC = 36^\circ + 38^\circ = 74^\circ$ (1); $\angle ADC = 180^\circ - 74^\circ = 106^\circ$ (1).	1+1
12	Distances: $\sqrt{5^2 - 3^2} = 4$ cm and $\sqrt{5^2 - 4^2} = 3$ cm (1); Distance $= 4 + 3 = 7$ cm (1).	1+1
13	Distances of centres from the chord: 9 cm and 16 cm (1); Distance $= 25$ cm (1).	1+1
14	$\angle ABC = 45^\circ + 22^\circ = 67^\circ$ (1); $\angle ADC = 180^\circ - 67^\circ = 113^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$d_1 = \sqrt{10^2 - 8^2} = 6$ cm, $d_2 = \sqrt{17^2 - 8^2} = 15$ cm (1); Opposite sides: $6 + 15 = 21$ cm (1); Same side: $\lvert 6 - 15 \rvert = 9$ cm (1).	1+1+1
16	Distances from the centre 8 cm and 6 cm (1); Same side: 2 cm (1); Opposite sides: 14 cm (1).	1+1+1
17	$\angle BCD = 180^\circ - 68^\circ = 112^\circ$ (1); $\angle ADB = 90^\circ$ (angle in a semicircle) (1); $\angle ABD = 90^\circ - 68^\circ = 22^\circ$ (1).	1+1+1
	OR: $d_1 = \sqrt{17^2 - 15^2} = 8$ cm, $d_2 = \sqrt{25^2 - 15^2} = 20$ cm (1); Opposite sides: $8 + 20 = 28$ cm (1); Same side: $\lvert 8 - 20 \rvert = 12$ cm (1).	1+1+1
18	Distances from the centre 4 cm and 3 cm (1); Same side: 1 cm (1); Opposite sides: 7 cm (1).	1+1+1
	OR: $\angle BCD = 180^\circ - 76^\circ = 104^\circ$ (1); $\angle ADB = 90^\circ$ (angle in a semicircle) (1); $\angle ABD = 90^\circ - 76^\circ = 14^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\angle D = 180^\circ - 95^\circ = 85^\circ$ (1). (ii) $(2x + 10) + (3x - 10) = 180 \Rightarrow x = 36$ (1). (iii) $\angle A = 82^\circ$, $\angle C = 98^\circ$ (1). (iv) $\angle DCE = 180^\circ - 98^\circ = 82^\circ$ (1), which equals $\angle A = 82^\circ$ because $\angle A + \angle C = 180^\circ$ (1).	1+1+1+2
20	$\angle A + \angle C = 180^\circ$ (1); $3x = 135$, so $x = 45$ (1); $\angle B + \angle D = 180^\circ$ (1); $5y = 155$, so $y = 31$ (1); $\angle A = 75^\circ$, $\angle C = 105^\circ$, $\angle B = 113^\circ$, $\angle D = 67^\circ$ (1).	1+1+1+1+1
	OR: Half-chord $= 10$ cm, $d = \sqrt{26^2 - 10^2} = 24$ cm (1); Area $= \frac{1}{2} \times 20 \times 24 = 240$ cm^2 (1); $OA = OB = 26$ cm (1); Perimeter $= 26 + 26 + 20 = 72$ cm (1); Distance, area and perimeter obtained (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).

- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND